ABDULLA JASEM ALMANSOORI

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RESEARCH SUMMARY

My research focuses on efficient collaborative learning via low-rank adaptation. I develop and analyze methods for personalized federated learning and distributed optimization, motivated by privacy and communication constraints, especially in healthcare and other sensitive, scarce data settings.

EDUCATION

Mohamed bin Zayed University of AI (MBZUAI) — Abu Dhabi, UAE
PhD in Machine Learning (Supervisor: Martin Takáč & Samuel Horváth) — (Aug 2021 - May 2026)

University of Southern California (USC) — Los Angeles, CA
MS in Computer Science — (Jan 2017 - Dec 2018)

Purdue University — West Lafayette, IN
BS in Industrial Engineering (with distinction) — (Aug 2012 - May 2016)

WORK EXPERIENCE

RIKEN – Center for Advanced Intelligence Project (AIP) — Tokyo, Japan
Research Intern (PI: Emtiyaz Khan & Thomas Möllenhoff) — (Aug 2025 - Feb 2026)

· Worked in the Adaptive Bayesian Intelligence team. Developed accelerated methods for variational learning that improved convergence and robustness relative to the lab baseline (IVON).

Meta – Monetization: Ranking and AI Foundations — Sunnyvale, CA
ML Engineer Intern (Mentor: Fan Xia) — (May 2025 - Aug 2025)

· Built 20M+ sample training pipelines for ads ranking, trained 7B-parameter multimodal models, and distilled 1B-parameter models for production use, improving offline AUC vs. baseline.

Sheikh Shakhbout Medical City (SSMC) — Abu Dhabi, UAE
Trainee (part-time) (Mentor: Siddiq Anwar) — (Jan 2025 - Present)

· Leading a project on applying federated learning in healthcare for 30-day readmission prediction with collaboration across multiple hospitals in the UAE. IRB approved and data ready, study in progress.

UAE National Service, Dubai Police — (Feb 2021 - Jun 2022)

· Finished army boot camp. Worked as a security guard in Dubai International Airport and Al Maktoum International Airport and as a vehicle inspector in Expo 2020.

New York University – Abu Dhabi (NYUAD) — Abu Dhabi, UAE
Research Assistant (PI: Kris Gunsalus) — (Jan 2020 - Feb 2021)

· Worked in the Center of Genomics and Systems Biology with the core bioinformatics team. Supported next-generation sequencing workflows and AI-based bioinformatics pipelines.

PUBLICATIONS

Preprint

1. *Beyond SGD, Without SVD: Proximal Subspace Iteration LoRA with Diagonal Fractional K-FAC.* (2026)
Abdulla Jasem Almansoori, Maria Ivanova, Andrey Veprikov, Aleksandr Beznosikov, Samuel Horváth, and Martin Takáč. arXiv preprint.

Conference

1. *Collaborative and Efficient Personalization with Mixtures of Adaptors.* (2025)
Abdulla Jasem Almansoori, Samuel Horváth, and Martin Takáč. Conference on Parsimony and Learning (CPAL).

2. *Byzantine-Tolerant Methods for Distributed Variational Inequalities.* (2023)
Nazarii Tupitsa, **Abdulla Jasem Almansoori**, Yanlin Wu, Martin Takáč, Karthik Nandakumar, Samuel Horváth, Eduard Gorbunov. Neural Information Processing Systems (NeurIPS).

Journal

1. *PaDPaF: Partial Disentanglement with Partially-Federated GANs.* (2024)
Abdulla Jasem Almansoori, Samuel Horváth, and Martin Takáč. Transactions on Machine Learning Research (TMLR).

2. *Stochastic Gradient Methods with Preconditioned Updates.* (2024)
Abdurakhmon Sadiev, Aleksandr Beznosikov, **Abdulla Jasem Almansoori**, Dmitry Kamzolov, Rachael Tappenden, and Martin Takáč. Journal of Optimization Theory and Applications (JOTA).

Workshop

1. *Faster Than SVD, Smarter Than SGD: The OPLoRA Alternating Update.* (2025)
Abdulla Jasem Almansoori, Maria Ivanova, Andrey Veprikov, Aleksandr Beznosikov, Samuel Horváth, and Martin Takáč. OPT 2025 Workshop.

2. *PaDPaF: Partial Disentanglement with Partially-Federated GANs.* (extended for TMLR) (2023)
Abdulla Jasem Almansoori, Samuel Horváth, and Martin Takáč. FLSys 2023 Workshop (oral).

POSTERS & TALKS

1. *Schedule-Free Variational Newton Methods.* (Feb 2026)
Talk, internal seminar, RIKEN-AIP.

2. *Collaborative and Efficient Personalization with Mixtures of Adaptors.* (Nov 2024)
Poster, ADIA Lab Symposium 2024.

3. *Federated Personalization with Mixtures of LoRAs.* (Dec 2023)
Poster, 2nd Collaborative Learning Workshop, MBZUAI.

4. *Lower Bounds for Non-Convex Stochastic Optimization.* (Oct 2022)
Talk, optimization seminar, MBZUAI.

ACADEMIC SERVICE

- **Teaching Assistant, MBZUAI**: (i) ML701: Machine Learning (Fall 23, Fall 24), (ii) ML802: Advanced Topics in Machine Learning (Spring 24), (iii) ML814: Selected Topics in Machine Learning (Spring 25), (iv) ML8509: Collaborative Learning (Spring 26).
- **Reviewer**: (i) NeurIPS 2022-2025, (ii) ICML 2023-2026, (iii) ICLR 2023-2025, (iv) ICOMP 2024, (v) TMLR.

AWARDS & HONORS

- **ICML Best Reviewer** Award, 2024.
- **MBZUAI Executive Program** Project Coordinator, 4th Cohort (2023).
  - Coordinated an AI strategy project for the government with executive-level participants.
- **Kawader Research Assistantship**, NYUAD, 2020.
- **Phi Beta Kappa** Society, Purdue University, 2016.

- **Distinguished Students Scholarship**, UAE Scholarships Office (SCO), 2012.
  - UAE's most prestigious academic scholarship.

SKILLS

Programming Languages	Python, C, JS, Haskell
ML/Research	PyTorch, Hugging Face, federated learning (flwr)
Experimentation/Systems	Hydra, Slurm, Docker, Git
Languages	Arabic (native), English (fluent), Japanese (intermediate), Italian (intermediate)